Matthieu Chaldebas | mchaldebas.github.io | Matthieu.Chaldebas@gmail.com | Publications

Computational biologist working at the intersection of machine learning and human infectious-disease genetics. Built 5ULTRA, a variant-effect prediction framework (*AJHG*, 2026). Experienced in designing and validating predictive-model evaluations at population scale (490,000 genomes).

EDUCATION

Université Paris Cité | PhD Candidate, Bioinformatics — 2022 – 2026 (Expected)

Sup' Biotech | MS in Biotechnology Engineering — 2016 – 2021

UC San Diego (UCSD) | Exchange Semester: Immunology & Human Disease — 6 months in 2018

EXPERIENCE

The Rockefeller University, New York, NY → UT Southwestern, Dallas, TX — 2021 – 2026
Laboratory of Human Genetics of Infectious Diseases, Casanova Lab
PhD Research & Bioinformatics Analyst

- Developed 5ULTRA, a **Random Forest ML framework** for functional variant prioritization; outperformed the industry-standard CADD score in AUROC/AUPRC on test datasets. (published in *AJHG* in 2026).
- Evaluated the framework: how much of a competing tool's performance came from the **mechanism** being modelled, genomic-inflation **calibration** across 125,754 tests, and covariate-matched null **backgrounds** for enrichment claims.
- Led the functional **analysis of the 490,640 UK Biobank** (UKB) whole genomes using 5ULTRA; identified novel rare-variant associations overlooked by the original consortium's approach (Preprint, under review).
- Orchestrated high-throughput pipelines to process and annotate **26,000+ WES/WGS** datasets using High-Performance Computing and SLURM, managing multi-terabyte genomic data flows.
- Validated ML predictions using **UKB Olink proteomics data** (n=50,000).
- Conducted **bulk and single-cell RNA-seq analyses**, enhancing transcriptomic insights across various projects; collaborate with a multidisciplinary team to design and implement custom analytical pipelines.

BioMérieux, Lyon, France — 6 months in 2021
Data Scientist Intern

- Optimized supervised **Lasso ML models** to increase predictive accuracy for antibiotic resistance.
- Enhanced **cross-validation** to account for phylogenetic factors, improving model generalization.
- Implemented algorithms for novelty detection in heterogeneous datasets.

CEA, Paris Saclay, France — 4 months in 2019
Data Engineer Intern

- Managed biological datasets, performed phylogenetic analysis, and created **3D protein models.**

CONFERENCES

ASHG 2026 in Montréal	Platform presentation on the UK Biobank study
ISMB 2026 in Washington DC	Tech Track Presentation
ASHG 2025 in Boston	Platform presentation on 5ULTRA

SKILLS

Computation: Python, Snakemake, R, Bash, Docker, Git, UKB RAP, DNAnexus, HPC (SLURM).
Machine-Learning: Scikit-learn, TensorFlow, Model selection, Feature Engineering, AlphaGenome Integration.
Statistical Genetics: Burden Testing, pQTL/Proteomics, WGS/WES Discovery, bulk/scRNA-seq, 5'UTR Biology.

Languages: French (Native), English (Fluent).